GUNDAGANI SATWIK

B.Tech + M.Tech
Department of Mathematics and Computing
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Summary

Software Development Enthusiast with strong fundamentals in data structures, algorithms, and backend development. Hands-on experience building scalable applications using Python, Flask, SQL, and modern web technologies and also in building AI-driven applications, Passionate about applied AI, LLM-based analytics, and full-stack dashboards for real-world decision-making.

Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech + M.Tech, Mathematics and Computing	National Institute of Technology, Patna	8.12/10 (till date)	2023 – 2028
B.S. in Data Science (Online)	Indian Institute of Technology, Madras	7.75/10 (till date)	2023 – Present
Senior Secondary	Sri Chaitanya Junior College	97.9%	2023
Secondary	Johnson Grammar School	80%	2021

Experience

- **Infosys Springboard Internship - AI Based Cybersecurity Threat Prediction Platform** *Sept 2025 – Nov 2025*
  Virtual Intern
  - Developed a hybrid Intrusion Detection System (IDS) fusing Classical ML (Random Forest, XGBoost) and Deep Learning (CNN-LSTM, Autoencoders) to detect network attacks, malware, and phishing with 95.3
  - Processed 3.4M+ security samples by integrating 2.8M CICIDS2017 logs, 500K Malware PE files, and 100K PhishTank URLs to identify DDoS, PortScans, and 8 diverse malware families.
  - Implemented a multi-layered architecture: used CNN-LSTMs for sequential byte-pattern analysis of malware, and Unsupervised Autoencoders with Isolation Forest to identify zero-day anomalies with an 89
  - Integrated an LLM-powered explainability layer to generate natural-language threat explanations and assist security decision-making.
  - Optimized the tech stack using Python, TensorFlow, and Scikit-learn for model development, used FastAPI with Joblib serialization for low-latency threat response.

Projects

- **Modern Application Development II: Vehicle Parking App – V2**
  Flask, Vue.js, Redis, Celery, SQLite, REST APIs
  - Developed a multi-user 4-wheeler parking management system with separate Admin and User roles, supporting multiple parking lots, dynamic spot allocation, and role-based dashboards.
  - Implemented Flask REST APIs, Vue.js frontend, Bootstrap UI, and SQLite schema for users, lots, spots, and reservations, with Redis caching to speed up frequent queries.
  - Built admin features for creating and managing parking lots, auto-generating spots, viewing occupancy, and monitoring registered users with summary charts.
  - Added Celery-based scheduled jobs for daily Google Chat/email reminders, monthly HTML/PDF activity reports, and user-triggered asynchronous CSV exports with completion notifications.

Skills

- **Programming Languages**: Python, SQL, C, Java
- **Machine Learning**: Pandas, NumPy, classical ML models, NLP
- **Web & APIs**: Flask, REST APIs, Vue.js, Frontend integration, Database design and management
- **Tools**: Jupyter, Git, Tableau, Redis, Celery, MySQL, Linux
- **Core CS**: Data Structures and Algorithms, Object-Oriented Programming, problem solving, Computer Networks, Operating Systems,

Certifications

- **Infosys Springboard 6.0**, Virtual Internship in AI-based Cybersecurity Threat Prediction Platform
- **Deloitte**, Data Analytics Virtual Job Simulation (Forage) *2025*
- **Eduxlabs, NIT Patna**, AI/ML with Python *2024*
- **KLUDGE, IIT Hyderabad**, AI/ML with Data Science *2024*

Achievements

- Ranked in the top 2% of candidates in JEE Mains 2023.
- Consistently maintained strong academic performance with 8.12 CGPA in dual-degree Mathematics and Computing and 7.75 CGPA in IIT Madras B.S. in Data Science.
- Qualified in Internal Hackathon for Smart India Hackathon 2024.

Positions of Responsibility

- **Class Representative**, B.Tech + M.Tech Mathematics and Computing, NIT Patna *2023 – Present*
- **Volunteer**, SUTRA Meetup, IIT Madras *2024*